Applied Soft Computing Journal 186 (2026) 114194

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

MultiDampGen: A self-constraint latent diffusion framework for multiscale energy-dissipating microstructure generation

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HIGHLIGHTS

- A dataset comprising 50,000 distinct damping microstructures and hysteresis characteristics is established.
- A multiscale damping microstructure generation (MultiDampGen) framework is proposed.
- MultiDampGen maintains low computational complexity while ensuring high generation accuracy.

ARTICLE INFO

Keywords:
Deep learning
Latent Diffusion
Multiscale
Inverse design
Damping microstructures

ABSTRACT

This study proposes MultiDampGen, a physics-informed latent diffusion framework that generates multiscale, shear-dominated damping microstructures aimed at seismic energy dissipation. The framework integrates a topology transformer (TopoFormer) for attention-enhanced latent compression, a residual-based self-constraint validator (RSV) for direct hysteresis prediction, and a latent diffusion physics mapper (LDPM) for conditional synthesis. A dataset comprising 50,000 finite-element–computed hysteresis curves under cyclic shear loading was constructed as the foundation of the proposed framework. Experimental results show that TopoFormer achieves a structural similarity index (SSIM) of 0.998 while reducing generative complexity by over 90 %, and RSV predicts complete hysteretic responses with an R^2 exceeding 0.98. MultiDampGen can generate microstructures meeting specified mechanical performance and scale requirements with deviations within 10 %, even under conflicting constraints such as a large scale combined with low load-bearing capacity. The framework enables efficient exploration of non-intuitive yet physically consistent designs, advances the application of generative artificial intelligence in earthquake engineering, and offers new insights for the development of next-generation energy dissipation systems.

1. Introduction

Seismic energy-dissipating design is crucial for improving the resilience of buildings. Post-earthquake investigations [1–6] have consistently highlighted insufficient energy-dissipating capacity as a major factor in structural collapse. Conventional metallic yielding [7–10] and viscous damping [11–13] technologies enhance seismic performance but often conflict with non-structural components, especially infill walls, in retrofit projects [14–16].

The energy-dissipating steel plate wall (EDSPW) [17] offers an innovative retrofit solution through its space-efficient thin-walled configuration [18], which enables surface-mounted installation on existing infill walls with negligible spatial intrusion while preserving architectural functionality, demonstrating significant advantages over conventional energy dissipation retrofitting approaches, as shown in Fig. 1. Typical EDSPW implementations utilize bidirectional arrays of engineered unit cells, a design strategy that aligns with metamaterial engineering principles where research-confirmed mechanical properties such as hyperelastic stiffness, negative Poisson's ratio, and exceptional strength-to-density ratios exceed those achievable with natural materials [19]. These mechanically optimized configurations exhibit superior ductility and energy-dissipating capacity compared to traditional systems. Unlike conventional compression-based metamaterial energy absorber typically employed for impact resistance [20–22], EDSPWs develop shear-governed energy-dissipating mechanisms when subjected to seismic cyclic loading, necessitating specialized design approaches

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https://doi.org/10.1016/j.asoc.2025.114194
Received 28 June 2025; Received in revised form 17 September 2025; Accepted 28 October 2025
Available online 6 November 2025
1568-4946/

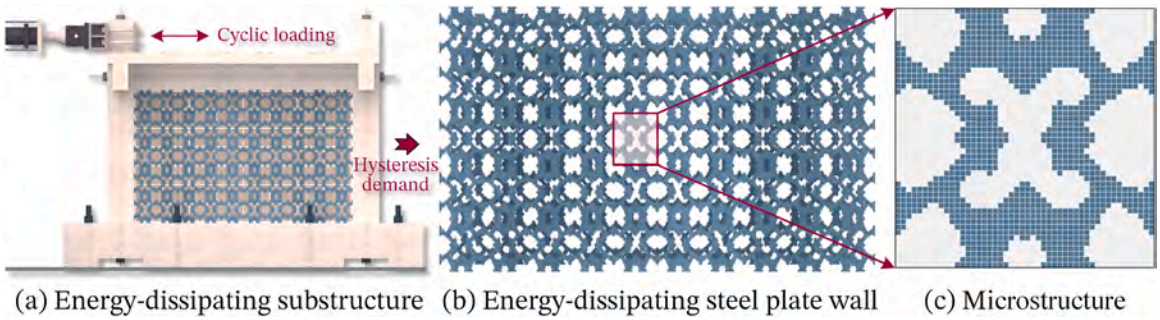

Fig. 1. Energy-dissipating structure and damping microstructure.

distinct from those used in compression-dominated systems.

Designing shear-governed damping microstructures is challenging due to the strong nonlinearities under earthquake loading [18]. Existing analytical approaches often fail under large deformations, while current generative AI methods [23–25], face three key issues: high computational cost [24], deviation from target mechanical properties, and limited single-scale generation capability. This study proposes MultiDampGen, a latent diffusion framework integrating a topology transformer, a residual-based self-constraint validator, and a physics-constrained generator to efficiently produce multiscale shear-dominated damping microstructures for EDSPWs. The main contributions are:

- A multiscale generative framework enabling synthesis of damping microstructures, with modular assembly capabilities that extend EDSPW applications for building seismic demands;
- Implementation of latent Diffusion models that substantially reduce computational resource consumption compared to conventional generative methods in EDSPW synthesis;
- Self-constraint optimization mechanisms addressing mechanical performance sensitivity to microscale geometric variations, ensuring compliance with structural hysteresis demands.

2. Related work

The primary challenge in the energy-dissipating microstructure reverse design lies in establishing a dataset that correlates images with mechanical performance data, as well as generating microstructures that meet specific conditions based on hysteresis performance and scale demands. Previous intelligent design methods have inadequately accounted for the variations in mechanical properties resulting from changes in microstructure scale, thereby limiting their practicality. This study provides a comprehensive summary of the research from two perspectives: the mechanical property dataset and vision generative models.

2.1. Mechanical property dataset for damping microstructures

Damping microstructures possess interconnected characteristics, which enable the formation of larger energy-dissipating components through periodic arrays. Consequently, the primary objective is to establish a dataset of interconnected structures. Following this, Finite Element Analysis (FEA) will be conducted on the generated microstructures to obtain the corresponding mechanical performance data [26–28]. The common methodology [25,29,30] for dataset construction via topological optimization involves iterative adjustments of material topology, filtering radius, and penalty factor initialization, followed by binarization processing of microstructural configurations. However, the microstructures produced by these methods are often excessively refined, leading to extreme deformations in energy-dissipating components under seismic loading, which may result in convergence difficulties during FEA. To address the issue of large deformations in metamaterial unit cells composed of non-uniform media, establishing benchmark datasets inspired by concepts from the field of computer vision presents a viable approach [31,32]. This methodology employs commonly used digits or letters as shapes for the metamaterials, facilitating the exploration of their mechanical performance through finite element analysis or enabling reverse design. However, such datasets frequently lack diversity, as the patterns tend to be simplistic and the fundamental structures of different unit cells are often too similar. Furthermore, the inadequate connectivity within these structures complicates the formation of periodic interconnected configurations. The methodology involving threshold sampling within two-dimensional Gaussian random fields [24] enables the generation of datasets containing geometrically intricate topologies with a wide distribution of stress-strain responses, and allows for the control of volume fraction within the resulting dataset. Overall, while the aforementioned methods have successfully established datasets that explore the tensile and compressive properties of microstructures, they have not addressed the shear-dominated, strongly nonlinear hysteresis characteristics that arise under cyclic seismic loading — a mechanical regime fundamentally different from the deformation modes typically considered in existing diffusion-based metamaterial generation studies. This gap is particularly critical in seismic engineering, where EDSPWs operate under such shear-dominated conditions. Furthermore, the scale of the datasets in prior work remains relatively fixed, which limits applicability across multiscale design demands.

In this study, we construct and openly release a large-scale seismic damping microstructure dataset containing over 50,000 samples. Each sample includes the geometry, physical scale, and hysteresis performance obtained from ABAQUS simulations, providing a rare and valuable resource for the research community (https://github.com/AshenOneme/MultiDampGen). This dataset forms the foundation of our physics-informed latent generative framework and supports accurate, efficient inverse design for seismic shear applications.

2.2. Vision generative models

Artificial Intelligence Generated Content (AIGC) refers to the capability of extracting and understanding intent information from instructions provided by humans, subsequently generating corresponding content based on this knowledge and intent. In recent years, significant achievements have been made in the fields of natural language processing, vision generation, and multimodal machine learning. As a critical area in generative modeling, manipulable image synthesis systems have witnessed a paradigm shift in methodological evolution. The technological trajectory has progressed from pioneering adversarial learning architectures, such as Generative Adversarial Networks (GANs) [23] and Variational Auto Encoders (VAEs) [33], to contemporary Diffusion models [34–36]. This evolution has demonstrated systematic enhancements in output fidelity, training convergence reliability, and

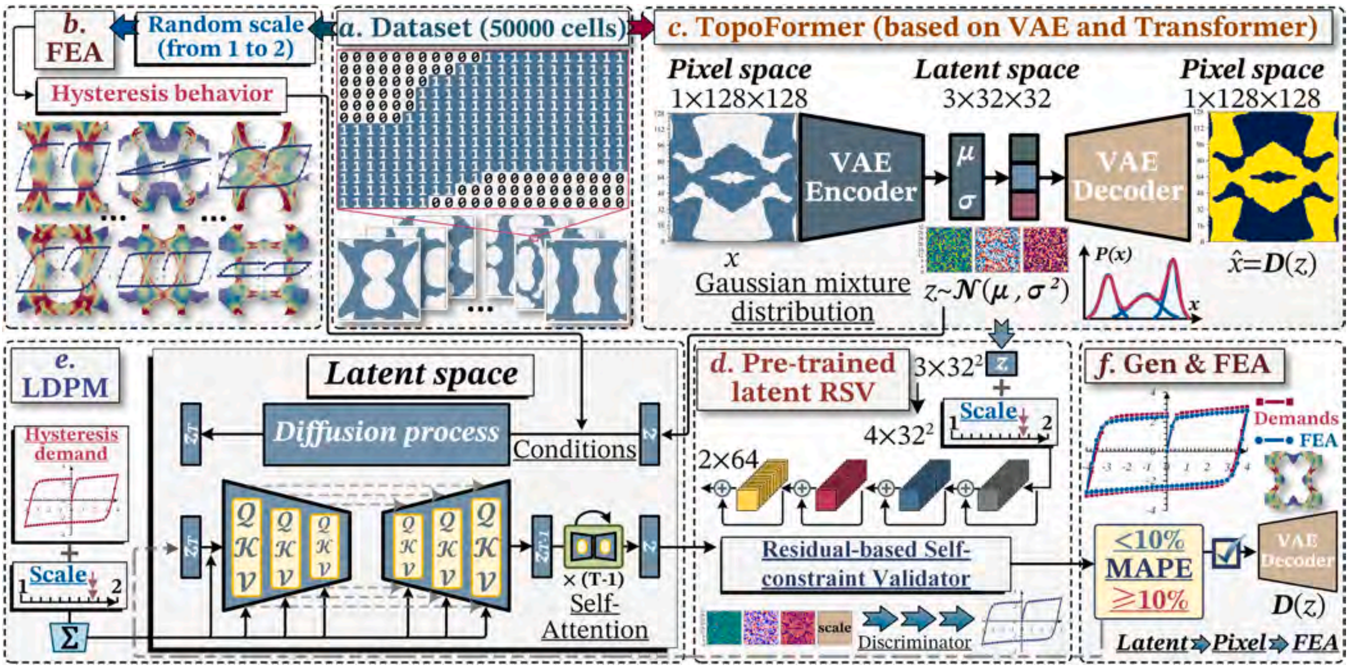

Fig. 2. The workflow of MultiDampGen framework.

parameterized controllability. The GANs employ an adversarial training mechanism to synthesize high-fidelity visual outputs; however, its operational scalability remains constrained by inherent training instability and mode collapse phenomena. In contrast, VAEs utilize a probabilistic inference paradigm to achieve stable optimization while constructing hierarchical latent representations, yet they exhibit lower perceptual accuracy in synthesizing image details. Diffusion models, as deep generative architectures rooted in nonequilibrium thermodynamic principles [37], have demonstrated unprecedented capabilities in contemporary machine learning research for synthesizing high-fidelity multimedia outputs [38] and modeling complex data distributions across various modalities [39].

Generative AI frameworks demonstrate unparalleled capacity for systematically navigating complex solution spaces, enabling the discovery of non-intuitive structural configurations and innovative material compositions that transcend conventional heuristic limitations in engineering design [27]. Whether employing VAEs [40] or GANs [41, 42], these computational frameworks demonstrate the capacity to synthesize microstructures conforming to predefined mechanical property constraint, while being capable of fabricating manufacturable configurations with counterintuitive morphological designs that satisfy ultra-stiff property requirements beyond the training distribution boundaries. Denoising Diffusion Probabilistic Models (DDPM) have emerged as a leading approach in the field of AIGC, demonstrating superior quality in content generation compared to previous methodologies. This technique has also found applications in the generation of microstructures [24,25,30,32], showcasing its versatility and effectiveness in producing high-quality designs. However, existing diffusion-based microstructure generation studies have predominantly targeted compression- or tension-driven mechanical behaviors with homogeneous materials and single-scale architectures, without incorporating the physics of seismic shear-dominated hysteresis into the generative process. In contrast, our MultiDampGen integrates a latent topology transformer, a physics-constrained validator, and a conditional latent diffusion mapper to directly generate multiscale microstructures that satisfy specified hysteresis curves under cyclic shear loading. This combination of application domain, mechanical regime, and physics-informed latent generation represents a distinct departure from prior works.

2.3. Novelty of this study

The above review clearly indicates that previous microstructure dataset construction methods have primarily focused on tension- or compression-driven mechanical behaviors, typically at a fixed scale, and that vision-based generative models have not incorporated the underlying physics of shear-dominated hysteresis phenomena into the design process. In contrast, the present study investigates the strongly nonlinear hysteresis characteristics of EDSPWs under cyclic seismic loading, an aspect largely overlooked in prior research. We establish an open, large-scale, multiscale dataset comprising 50,000 microstructures, each annotated with geometry, physical scale, and hysteresis performance. Based on this dataset, we propose MultiDampGen, a physics-informed latent diffusion framework that combines a topology transformer for efficient latent representation, a residual-based self-constraint validator for direct hysteresis prediction, and a latent diffusion physics mapper for conditional synthesis. This integrated approach enables accurate and efficient generation of multiscale, shear-dominated damping microstructures that meet specified mechanical performance requirements, thereby bridging a critical gap between general-purpose generative AI methods and the specialized demands of seismic energy dissipation design.

3. Methodology

3.1. Overview of the workflow

The MultiDampGen framework represents an integrated computational architecture consisting of four synergistic components: a methodology for mapping geometric image-to-FEA for cross-domain transformation, a microstructure Topology Transformer (TopoFormer) that facilitates multiscale topology compression through attention-based morphogenesis, a Residual-based Self-constraint Validator (RSV) that quantifies physics-aware discrepancies by analyzing latent space and scale information, and a Latent Diffusion Physics Mapper (LDPM) designed to generate latent space variables that satisfy hysteresis performance demands. This workflow systematically progresses through sequential phases from morphological parameterization to physics-informed performance validation, as illustrated in Fig. 2.

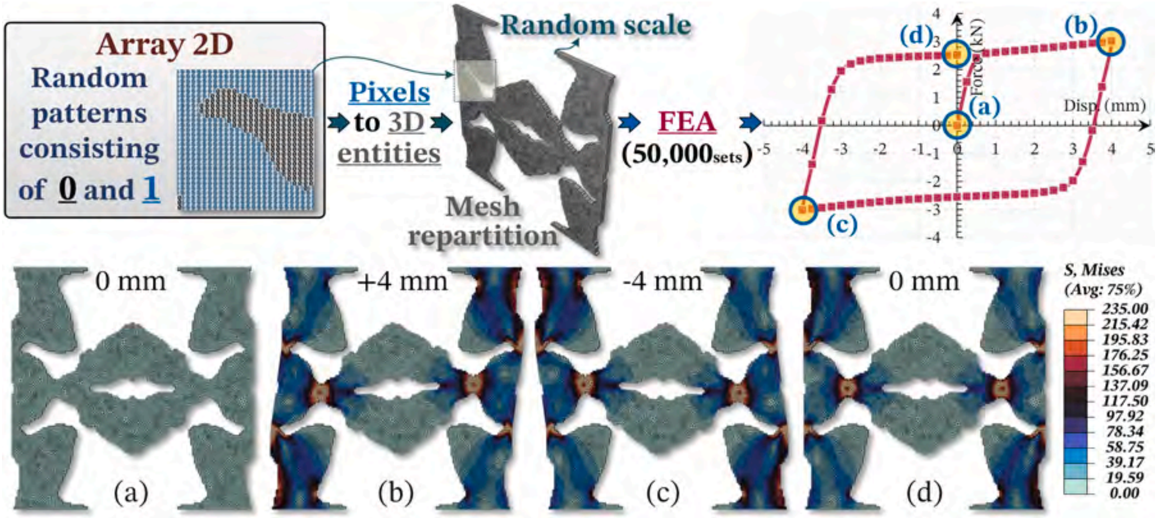

Fig. 3. The process of dataset establishment.

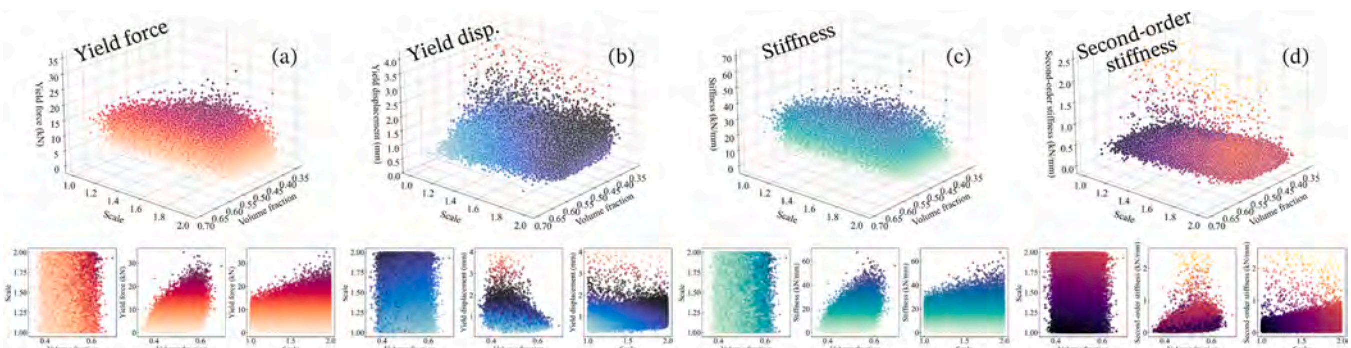

Fig. 4. Distribution of mechanical properties.

The MultiDampGen framework establishes a bidirectional mapping mechanism between pixel space representations and physical realizations while facilitating the translation of abstract mechanical performance specifications into tangible microstructural configurations. Its multiscale and self-constraint characteristics represent a significant departure from prior research [24,25,30,32]. The computational implementation of this architecture, which includes the core innovation of bridging data-driven morphology synthesis with physics-constrained performance validation, will be systematically elucidated in subsequent sections.

3.2. Multiscale microstructure hysteresis performance dataset

The energy-dissipating microstructure dataset requires the establishment of interconnected random patterns. Utilizing a Gaussian random field sampling method [24], a total of 50,000 random patterns were generated. Following binarization, each pattern was defined with dimensions of 128 × 128 pixels, and a random scale was assigned to each pixel. The physical scale was set between 1 and 2 mm, resulting in an actual physical size of each pattern ranging from 128 to 256 mm. A conversion plugin was developed using the ABAQUS-Python API to transform binary arrays into 3D geometric entities, with each entity was assigned a thickness of 4 mm and was constructed from steel with a yield strength of 235 MPa. After remeshing each 3D entity, a shear cyclic simulation was conducted, yielding a dataset comprising 50,000 sets of data corresponding to geometric shapes, pixel physical scales, and hysteresis performance. This formed the basis for the establishment of the final dataset. Fig. 3 illustrates the Mises stress cloud maps corresponding to four critical points of mechanical performance. During the automated modeling process, the random scale and patterns contributed to the independence of stress results for each microstructure.

The distributions of yield force, yield displacement, stiffness, and second-order stiffness associated with the multiscale microstructure hysteresis performance dataset are presented in Fig. 4. The horizontal axes in both directions represent the scale and volume fraction of each microstructure, while the subplots illustrate the pairwise relationships among the three coordinate axes. The scale is evidently uniformly distributed between 1 and 2 mm, indicating that the height and width of all microstructures are uniformly distributed within the range of 128–256 mm. The distribution of yield strength closely resembles that of stiffness, with both parameters exhibiting an overall increasing trend as the scale and volume fraction increase. However, this distribution is also characterized by considerable dispersion, as instances of low yield strength and stiffness can be found even at high volume fractions and scales. This observation highlights the significant variability among configurations within the microstructural dataset, suggesting that not all parts contribute equally to mechanical performance. Many ineffective portions appear to increase the dimensions of the microstructure without bearing load, a phenomenon that is also reflected in Fig. 3.

Analysis of four mechanical property distributions reveals statistically insignificant correlations between mechanical performance and geometric configurations in multiscale microstructures, as shown in Fig. 4. Specifically, specimens with divergent dimensions may demonstrate comparable mechanical characteristics, while geometrically analogous counterpart conversely exhibit substantial discrepancies in

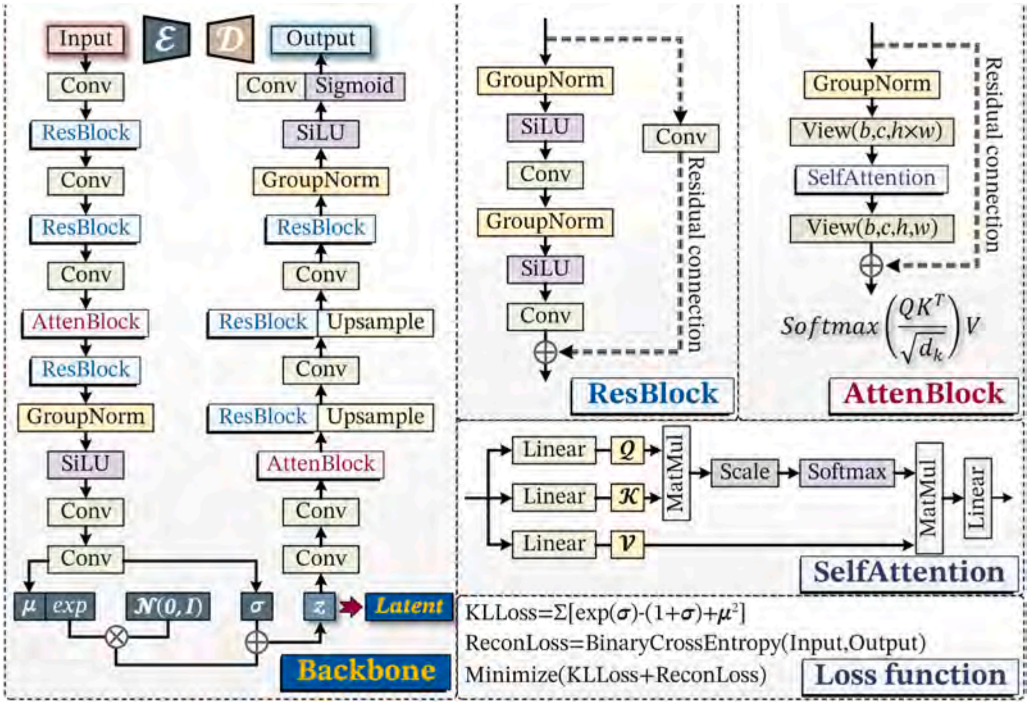

Fig. 5. Architecture of microstructure topology transformer.

mechanical performance. This statistical phenomenon underscores the critical necessity of implementing the RSV during microstructure generation to effectively decouple geometric variance from mechanical response characteristics.

3.3. TopoFormer

Following dataset construction, the implementation of a TopoFormer network for hierarchical feature extraction of microstructures constitutes the primary computational phase, as demonstrated in Fig. 2(c). This architecture draws inspiration from Latent Diffusion principles [43], particularly leveraging its spatial encoding mechanism to establish topology-descriptor mappings while maintaining geometric fidelity through constrained variational autoencoding. The TopoFormer can be expressed mathematically through the optimization of the following objective function:

$$\mathscr{L}_{total} = \mathbb{E}_{z\sim q(z|x)}[\mathscr{L}_{recon}] + \beta \bullet \mathscr{L}_{KL} \tag{1}$$

$$\mathscr{L}_{recon} = -\frac{1}{N}\sum_{i=1}^{N}(y_i \log(\widehat{y}_i) + (1-y_i)\log(1-\widehat{y}_i)) \tag{2}$$

$$\mathscr{L}_{KL} = \frac{1}{2}\sum_{i=1}^{d}\left(\sigma_z^{(i)} - \log\left(\sigma_z^{(i)}\right) - 1 + \mu_z^{(i)2}\right) \tag{3}$$

where x is the input microstructure data, z is the latent variable, N represents the number of samples, y_i denotes the true labels, $\widehat{y}_i$ indicates the predicted results by the model, $\sigma_z^{(i)}$ is the variance parameter of the posterior distribution for the latent space, and $\mu_z^{(i)}$ is the mean parameter. In the TopoFormer, the transformations performed by the encoder ($\mathscr{E}$) and decoder ($\mathscr{D}$) establish a nonlinear mapping between the pixel

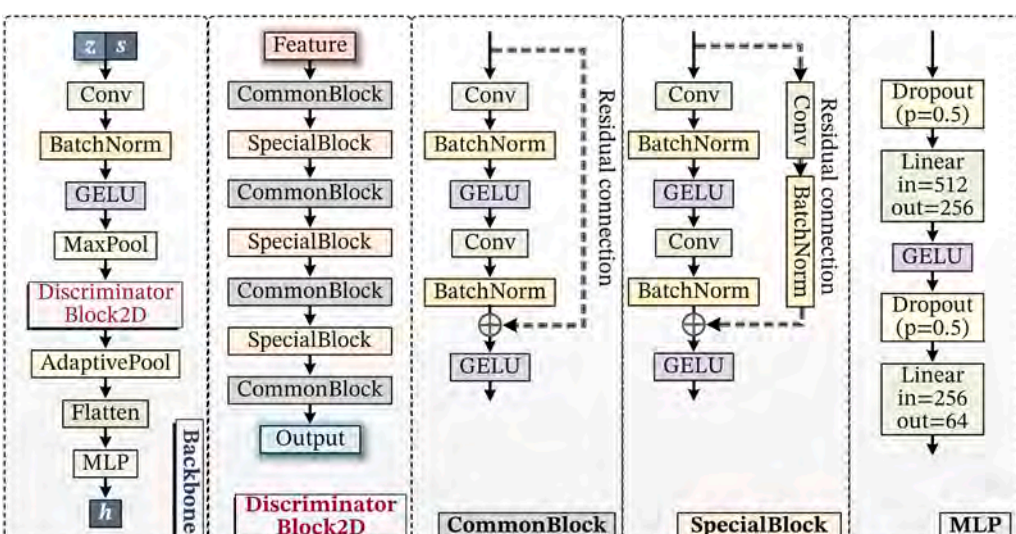

Fig. 6. Architecture of residual-based self-constraint validator.

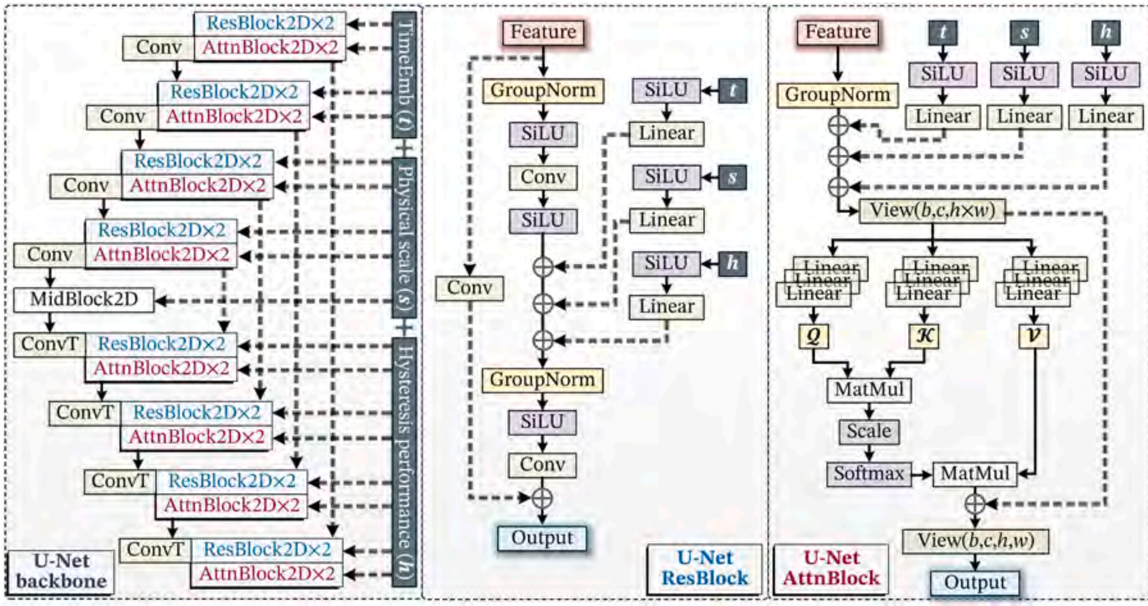

Fig. 7. Architecture of Latent Diffusion Physics Mapper.

space $x \in \mathbb{R}^{H\times W}$ and latent space $\mathscr{z} \in \mathbb{R}^{3c\times d}$ with $d = H/4$. The hierarchical feature contraction within the encoder stacks implements multi-scale geometric abstraction through strided convolutions:

$$H^{(l+1)} = H^{(l)} + Atten(Q, K, V) \tag{4}$$

where $\mathscr{M}_{attention}$ denotes the scaled dot-product attention [44] generated by:

$$Atten(Q, K, V) = Softmax\left(QK^T \middle/ \sqrt{d_k}\right)V \tag{5}$$

$$Q, K, V = Proj\left(H^{(l)}\right) \tag{6}$$

where Q, K and V are the query, key, and value matrices, respectively, and d is the dimension of the key vectors.

The overall architecture of TopoFormer is depicted in Fig. 5. Notably, during the latent variable generation phase, the output $\mathscr{z}$ undergoes nonlinear transformation through hyperbolic tangent activation (tanh) to constrain its values within the interval (-1, 1). Ultimately, the bounded latent representation tanh($\mathscr{z}$), serves as the foundation for training the LDPM and the RSV.

3.4. RSV

The RSV is employed to predict the hysteresis performance (h) of latent variables. This innovative approach utilizes latent representations ($\mathscr{z}$) enhanced by physical scale information (s), allowing for a more comprehensive understanding of the mechanical properties of microstructures. By integrating latent features and scale information, the RSV effectively captures the intricate relationships inherent within the data, enabling it to predict hysteresis behavior with high fidelity. For latent variables $\mathscr{z} \in \mathbb{R}^{3c\times d}$ and physical scale $\mathscr{z} \in \mathbb{R}^{c\times d}$, the hysteresis prediction h follows:

$$h = \mathscr{F}_{RSV}(\mathrm{Concat}(\tanh(\mathscr{z}), s)) \tag{7}$$

where $\mathscr{z} \sim \mathscr{N}(\mu, \sigma^2)$, $s \sim U(1, 2)$ and h is the finite element model calculated results.

The architecture of the RSV, as illustrated in Fig. 6, adopts a canonical residual network [45] structure. Given that microstructures often contain intricate substructures where detailed features significantly influence the hysteresis performance h, the discriminator block within RSV is intentionally designed without dimensionality-reduction operations such as pooling layers. This configuration ensures that the spatial dimensions remain unchanged throughout the network, thereby mitigating potential precision degradation that might arise from feature compression during downsampling processes. The preservation of spatial resolution enables more accurate retention of critical microstructural details essential for h prediction. The internal architecture of the discriminator block is realized by stacking special blocks and common blocks, with the distinction between the two lying in the presence or absence of convolution and normalization operations during the residual connection process. The final output structure undergoes adaptive pooling before being flattened and passed into the fully connected layers to obtain the h. The trained RSV allows for the prediction of generated outcomes during the generation process, significantly enhancing the accuracy of the generated results.

3.5. LDPM

The architecture of the LDPM adheres to the architecture of Latent Diffusion models, where the generation process begins with an input of noise. In this framework, the s and the h serve as independent conditioning variables that inform the generation of the latent representation $\mathscr{z}$. The core backbone of the LDPM is structured as a U-Net [46] network, which is divided into downsampling and upsampling layers, as shown in Fig. 7. Each layer consists of two stacked residual layers (ResBlock2D) and two attention layers (AttnBlock2D), with the unique distinction that downsampling and upsampling are achieved through strided convolutions and transposed convolutions, respectively. In the U-Net architecture, each downsampling block receives input from the previous module, the embedding results corresponding to the timesteps (t), and the mappings of both s and h. Conversely, the upsampling blocks are designed to simultaneously accept the semantic information from the downsampling process.

In contrast to DDPM, the LDPM operates in a low-dimensional perceptual equivalent space rather than directly in pixel space, resulting in theoretical differences. LDPM enhances the foundational U-Net backbone through residual networks and cross-attention mechanisms, which are particularly effective for learning attention models across various input modalities. By employing two domain-specific physical encoders, the cross-modal conditions s and h are projected into intermediate representation $\tau_\theta(s)$ and $\psi_\theta(h)$. For each ResBlock2D at layer l:

Table 1
Comparison with representative past studies.

Study & Year	Mechanical regime	Scale capability	Physics integration	Dataset size & availability	Generative approach
Lee et al. (2022) [29]	Generic elastic props.	Single	None	9.9k–88k, mixed	VAE + GP + DPP
Bastek & Kochmann (2023) [24]	Nonlinear comp./shear	Single	None	~53k, open	Video diffusion
Wang et al. (2024) [25]	Energy-absorbing comp.	Single	None	~30k, closed	Conditional diffusion
Feng et al. (2025) [30]	Linear elastic modulus	Multi (boundary-identical)	None	> 100k× 16, open	Self-cond. diffusion
This work	Shear-dominated cyclic hysteresis	True multi (128–256 mm)	Physics-informed latent diffusion	~50k, open	TopoFormer + RSV + LDPM

$$\mathscr{R}_{z}^{(l)} = c_{z}^{(l)} + \mathscr{C}^{(l)}\left(c_{z}^{(l)} + \varphi_\theta(t_{emb}) + \tau_\theta(s) + \psi_\theta(h)\right) \tag{8}$$

where $c_{z}^{(l)}$ denotes the convolution operation applied to the latent variables, $\varphi_\theta(t)$ represents the projection corresponding to timestep t. The time embedding t_{emb} is defined as:

$$t_{emb} = \mathscr{W}_2 \bullet Swish\left(\mathscr{W}_1 \bullet \begin{bmatrix}\sin(t \bullet \omega_i) \\ \cos(t \bullet \omega_i)\end{bmatrix} + b_1\right) + b_2 \tag{9}$$

$$\omega_i = \exp\left(-\frac{i \bullet \ln(10000)}{d/2 - 1}\right) \tag{10}$$

where $Swish(x)$ is the self-gated activation function and d is the embedding dimension.

The LDPM forward Diffusion propagates latent representations through a conditional Markov chain with physical state awareness:

$$q(z_{1:T}|z_0) = \prod_{t=1}^{T} q(z_t|z_{t-1}, s, h) \tag{11}$$

each transition implements physics-informed noise injection:

$$q(z_t|z_{t-1}, s, h) = \mathscr{N}\left(z_t; \sqrt{1-\beta_t}\, z_{t-1}, \beta_t I\right) \tag{12}$$

During the reverse generation phase, the LDPM initiates the synthesis process by sampling unstructured noise from a predefined prior distribution. This initial state subsequently undergoes iterative refinement through a learned reverse Markov Chain, which progressively eliminates noise components while incorporating physical guidance from s and h. The entire microstructure formation procedure can be mathematically formalized as:

$$p_\theta(z_{t-1}|z_t, s, h) = \mathscr{N}(z_{t-1}; \mu_\theta(z_t, t, s, h), \Sigma_\theta) \tag{13}$$

$$\mu_\theta(z_t, t, s, h) = \frac{1}{\sqrt{\alpha_t}}\left(z_t - \frac{\beta_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(z_t, t, s, h)\right) \tag{14}$$

where ϵ_θ represents the noise prediction network that predicts the combined noise-physics residuals. $\alpha_t = 1 - \beta_t$ and $\overline{\alpha}_t = \prod_{s=1}^{t}\alpha_s$ correspond to the temporal scheduling coefficients. Based on microstructure-conditioning pairs, the LDPM is optimized through the following physics-informed denoising objective:

$$\mathscr{L}_{LDPM} = \mathbb{E}_{\mathscr{E}(x), s, h, \epsilon\sim\mathscr{N}(0,I), t}\left[\|\epsilon - \epsilon_\theta(z_t, t, s, h)\|_2^2\right] \tag{15}$$

4. Procedures of experiments

This section offers a comprehensive explanation of the experimental methods and design, including the hyperparameters used for training the model and the evaluation methods for the generated results, in order to thoroughly assess the model's performance in generating damping microstructures (Table 1).

Table 2
Environment configuration.

Category	Component	Specification
Hardware	CPU	AMD EPYC 9654 96-Core Processor
	GPU	2 ×NVIDIA 3090 GPU
	RAM	480 GB
Software	System	Windows 10 Professional version
	Python	3.12
	ABAQUS	2020
Environment	Pytorch	2.6
	CUDA	12.4

Table 3
Parameter settings.

Category	Parameter	Value
AdamW optimizer	Learning rate	0.001
	Weight decay	0
Lr scheduler	ExponentialLR	0.98
DDPM	Timesteps	1000
	$\beta_{schedule}$	Squared cosine
	β_{start}	0.0001
	β_{end}	0.02
DDIM	Timesteps	100
	Dynamic thresholding ratio	0.99

4.1. Experimental setup

The experiments conducted in this study were based on a Windows 10 Professional operating system. During the dataset creation process, finite element simulations were performed using ABAQUS 2020, and the deep learning model was trained in a PyTorch environment utilizing dual NVIDIA 3090 GPUs, as shown in Table 2. The generated dataset comprises a total of 50,000 samples, with 45,000 samples allocated for the training set, while the remaining 5,000 samples serve to validate the microstructure reconstruction fidelity and latent compression efficiency, assess the prediction accuracy of RSV, and can also be used as hysteresis performance and physical scale conditions for the LDPM.

4.2. Training process and parameter details

The MultiDampGen framework is divided into three components, namely TopoFormer, RSV, and LDPM, with optimization conducted using the AdamW algorithm. The initial learning rate is set at 0.001, and exponential learning rate decay is applied with a factor of 0.98, with all components trained for 200 epochs. Specifically, for the LDPM, the noise scheduler follows the DDPM, where the number of timesteps is set to 1000, the beta schedule is defined as squared cosine, with the β_{start} set at 0.0001 and the β_{end} at 0.02. During the inference phase, the noise scheduler employs the DDIM approach, with the number of timesteps set to 100 and the dynamic thresholding ratio configured at 0.99. The β values are set in accordance with those used in the DDPM, as presented in Table 3.

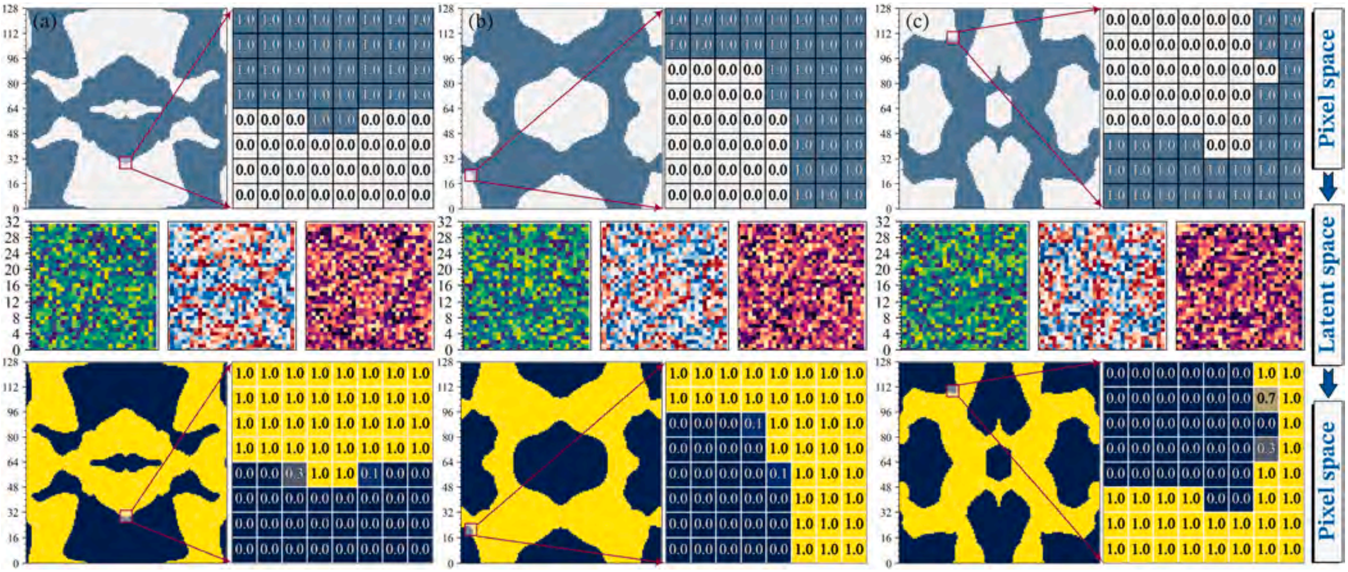

Fig. 8. Microstructure encoding and decoding.

To prevent overfitting and enhance generalization, the dataset was divided into 45000 training samples, 5000 validation samples, and a non-overlapping subset for independent testing. An early stopping criterion was applied when the validation loss failed to improve for 20 consecutive epochs, and an exponential learning rate decay was adopted to avoid overtraining. Dropout layers and weight decay regularization were incorporated into both RSV and LDPM to mitigate co-adaptation of parameters. In addition, data augmentation was performed through random scaling of physical dimensions, Gaussian noise injection in the latent space, and morphological perturbations of the microstructures. Collectively, these measures improve the model's robustness to unseen input distributions and reduce the risk of overfitting.

4.3. Evaluation method

In this study, the Structural Similarity Index Measure (SSIM) is employed to quantify the morphological consistency between the generated microstructures and all pre-existing microstructures within the dataset. The SSIM metric evaluates luminance, contrast, and structural similarity using a sliding window approach, and is defined as follows:

$$SSIM(x,y) = \frac{\left(2\mu_x\mu_y + C_1\right)\left(2\sigma_{xy} + C_2\right)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)} \tag{16}$$

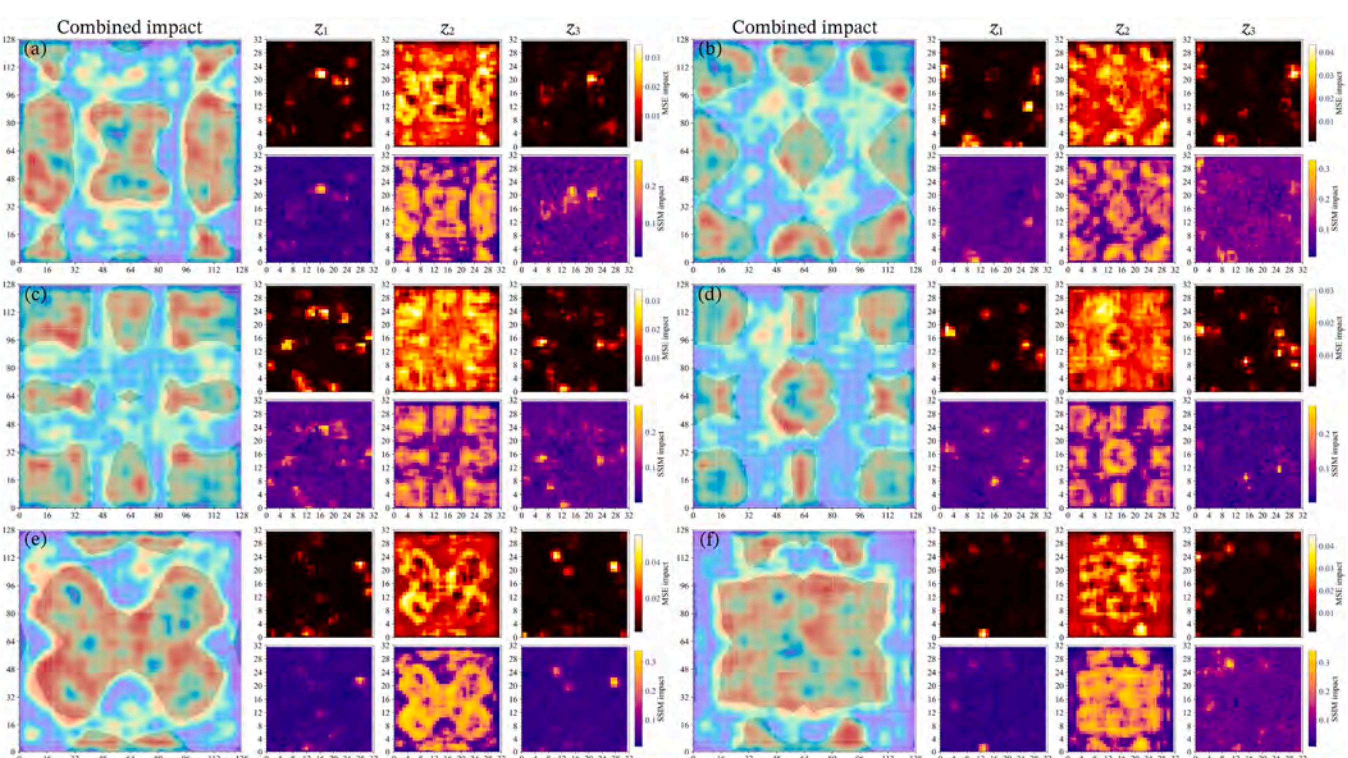

Fig. 9. Local sensitivity mapping heatmap.

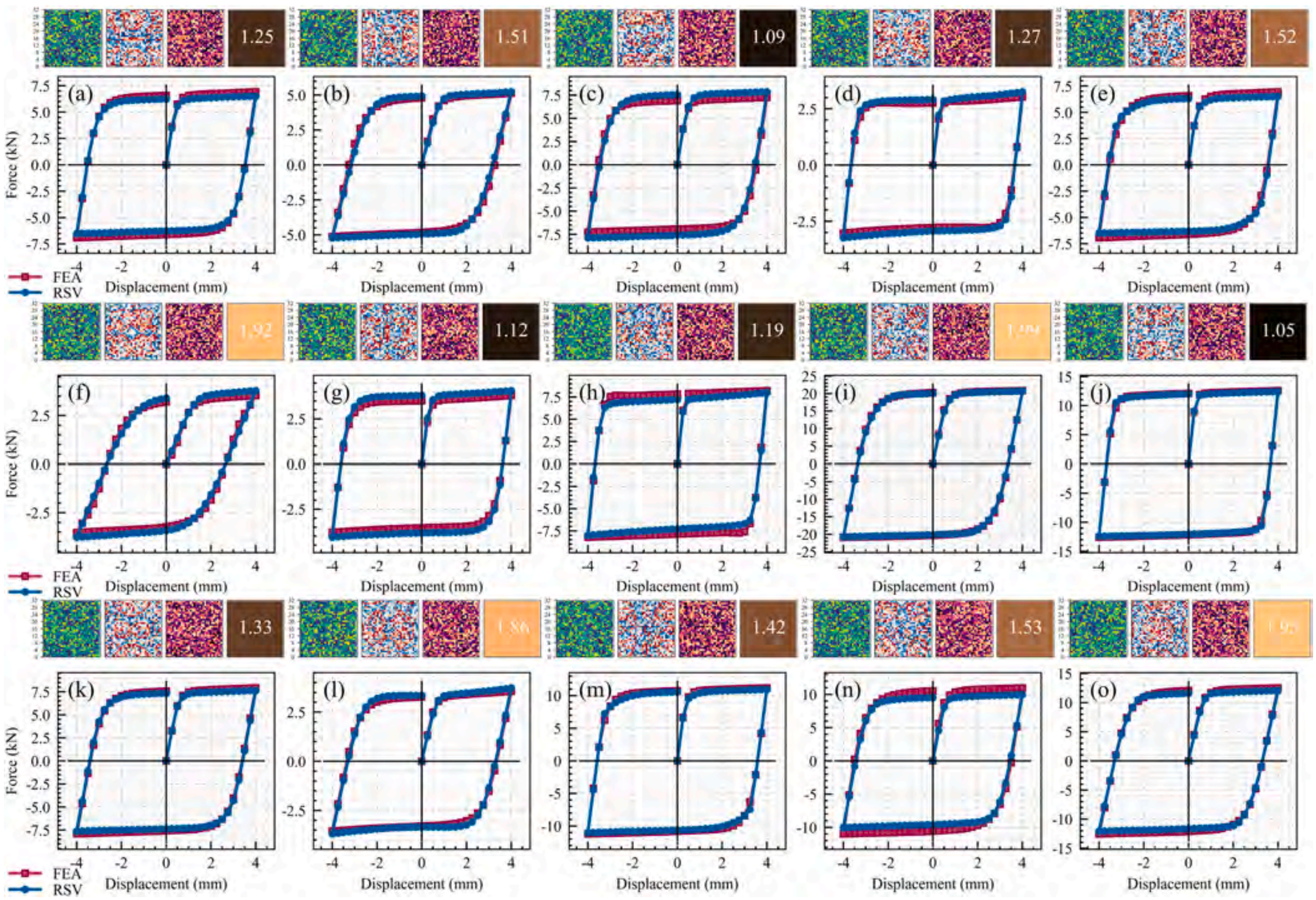

Fig. 10. The prediction results of RSV.

where μ_x and μ_y are the local means, σ_x^2 and σ_y^2 are the variances, σ_{xy} indicates the covariance of microstructure patches x and y. The benefit of utilizing SSIM lies in its capacity to evaluate whether the results generated by MultiDampGen represent entirely new configurations that have not been previously encountered or if they consist of configurations selected from the dataset that meet the demands for hysteresis performance.

5. Results and discussions of experiments

5.1. The microstructure reconstruction capabilities of TopoFormer

For different microstructures, TopoFormer employs an encoder to transform pixels into latent representations, followed by a decoder for reconstruction, as illustrated in Fig. 8. The similarity between the reconstructed results and the original microstructures is remarkably high, with the final model achieving a SSIM of 0.998 and a Root Mean Square Error (RMSE) of 0.022 in the test set. Benefiting from the robust compression and reconstruction capabilities of TopoFormer, the computational complexity of subsequent RSV and LDPM processes has been reduced by a factor of squared magnitude. Although TopoFormer demonstrates robust encoding and decoding capabilities, loss of precision still occurs at the corners of microstructures. In most cases, the generation errors at these corner locations do not significantly affect the mechanical properties of the microstructures. However, generation errors at critical locations may substantially increase or decrease the load-bearing capacity.

The analysis of latent space interpretability employs structured perturbation techniques to quantify the impact of different regions of the latent variables on the reconstruction results of microstructures. If significant changes in the reconstructed microstructure occur following perturbations in a specific region of the latent space, it indicates that this region encodes critical features; conversely, if minimal changes are observed, it may suggest the presence of redundancy or noise. The results of the impact analysis of the latent space are illustrated in Fig. 9. MSE is utilized to quantify pixel-level errors and capture high-frequency detail variations, while the SSIM is employed to assess structural similarity, reflecting characteristics of human visual perception. The results of the MSE impact and the SSIM impact are generally consistent. By employing a dual-metric approach, the limitations associated with a single metric are mitigated. The results of the latent space impact analysis indicate that the contribution of the latent variable z_2 to the reconstruction of microstructures is greater than that of z_1 and z_3. By employing bilinear interpolation to align the latent space impact heatmap with the original microstructure, a clear visualization of the mapping relationship between the latent space and the image semantics is achieved. Both z_1 and z_3 contribute significantly to the reconstruction details, and the overall reconstruction impact is largely in agreement with the original microstructure.

5.2. The predictive performance of RSV

All microstructures in the test set were transformed into latent variables using TopoFormer, and the prediction of hysteresis characteristics was conducted by incorporating the physical scale as a feature. The performance obtained through FEA is represented by the red curve, while the predictions generated by the trained RSV model are illustrated by the blue line, as shown in Fig. 10. The curves presented exhibit significant variation in terms of stiffness, load-bearing capacity, and energy

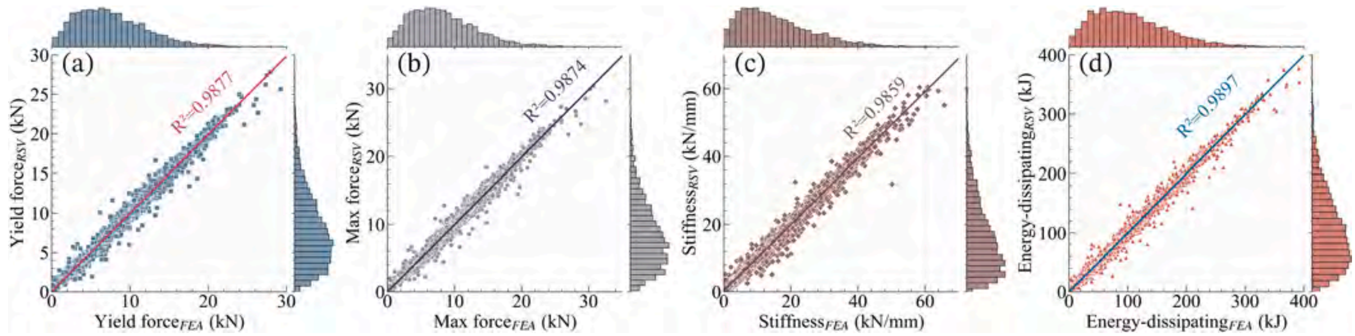

Fig. 11. Comparison of predictive hysteresis performance.

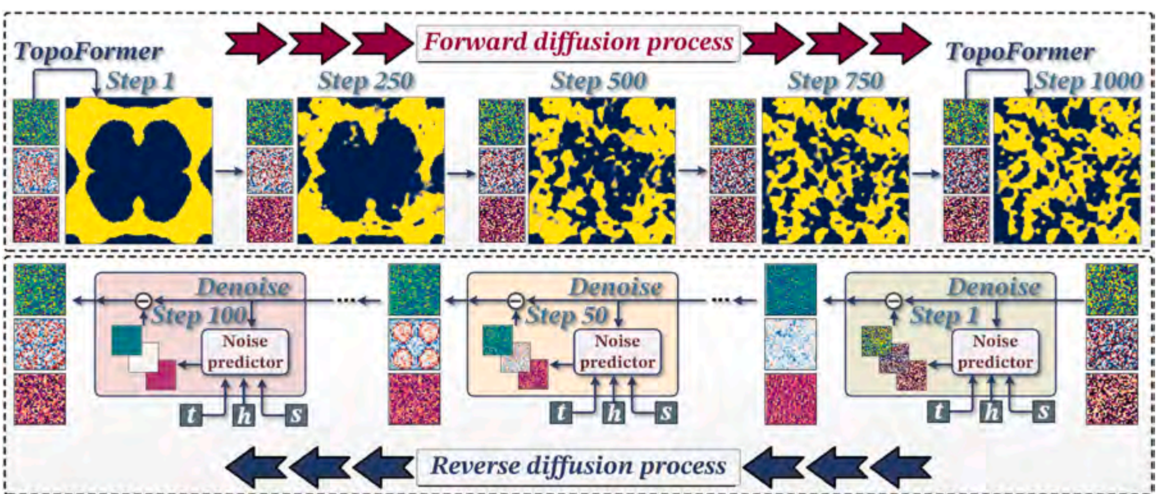

Fig. 12. The forward and inverse processes of LDPM.

dissipation capability, as illustrated in Fig. 10(f), where the physical scale is 1.92. In this case, the maximum load-bearing capacity is 3.79 kN, and the stiffness is 2.72 kN/mm. In contrast, the microstructure corresponding to Fig. 10(i) has a physical scale of 1.99, which is similar to that of Fig. 10(f), yet it demonstrates a substantially higher maximum load-bearing capacity of 20.73 kN and a stiffness of 3.96 kN/mm. This discrepancy highlights the significant imbalance present within the predicted dataset; however, the RSV model still maintains a relatively high level of accuracy in its predictions.

Unlike other AIGC tasks, the microstructure generation task imposes higher requirements on mechanical performance, which is not solely reflected in load-bearing capacity. Therefore, a comparative analysis of the prediction results from the RSV model is conducted in terms of yield force, maximum force, stiffness, and energy-dissipating capacity against the FEA results, as illustrated in Fig. 11. The four mechanical performance indicators exhibit a marked degree of extreme imbalance, with the performance of various microstructures differing by factors ranging from several tens to several hundreds across different metrics. Nevertheless, the predictive performance of the RSV model demonstrates considerable robustness, despite having a relatively small parameter count of only 1.13 million. This makes RSV particularly suitable as the self-constraint generative component of MultiDampGen, owing to its simplicity and low parameter count. It is worth noting that the RSV does not predict isolated scalar mechanical properties such as yield force or stiffness; instead, it predicts the entire hysteresis curve as a continuous response. This holistic target formulation substantially mitigates the impact of statistical imbalance in any single parameter, as the loss function is defined over the full cyclic loading trajectory rather than on point estimates. The R^2 for different indicators consistently exceed 0.98; however, accuracy tends to be relatively lower for larger and more extreme samples, which is attributed to the inherent imbalance in the training dataset.

5.3. The generation process of MultiDampGen

As a generation framework for multiscale damping microstructures, MultiDampGen relies on the perturbation and denoising of data through LDPM during its training and inference processes, as illustrated in Fig. 12. According to Eq. 15, noise is gradually added to the latent variable z, and during this process, it can be observed that the data structure remains disturbed when z is reconstructed back into pixel space through the TopoFormer decoder. By executing the trained Markov chain in reverse as outlined in Eq. 18, noise predictions are made, constituting the reverse denoising process. By imposing the constraints of the condition t, s, and h, microstructures that meet the demands for hysteresis performance can be generated. Ultimately, the results from the latent space are decoded into pixel space to construct the microstructures, which are then subjected to FEA computations to obtain the mechanical properties of the generated microstructures. It is noteworthy that both the forward and inverse processes involve operations conducted within the latent space. The LDPM comprises over 39.75 million parameters, while the TopoFormer performs encoding and extraction of data related to the microstructures. Consequently, MultiDampGen demonstrates a high inference efficiency.

Three hysteresis performance demand curves were randomly selected, with maximum force demands of 12.07 kN, 12.40 kN, and 3.79 kN, respectively. The corresponding s were 1.95, 1.05, and 1.92, indicating microstructure sizes of 249.60 mm, 134.40 mm, and 245.76 mm, as shown in Fig. 13. The mechanical performance demands depicted in Fig. 13(a) are similar to those in Fig. 13(b); however, there is

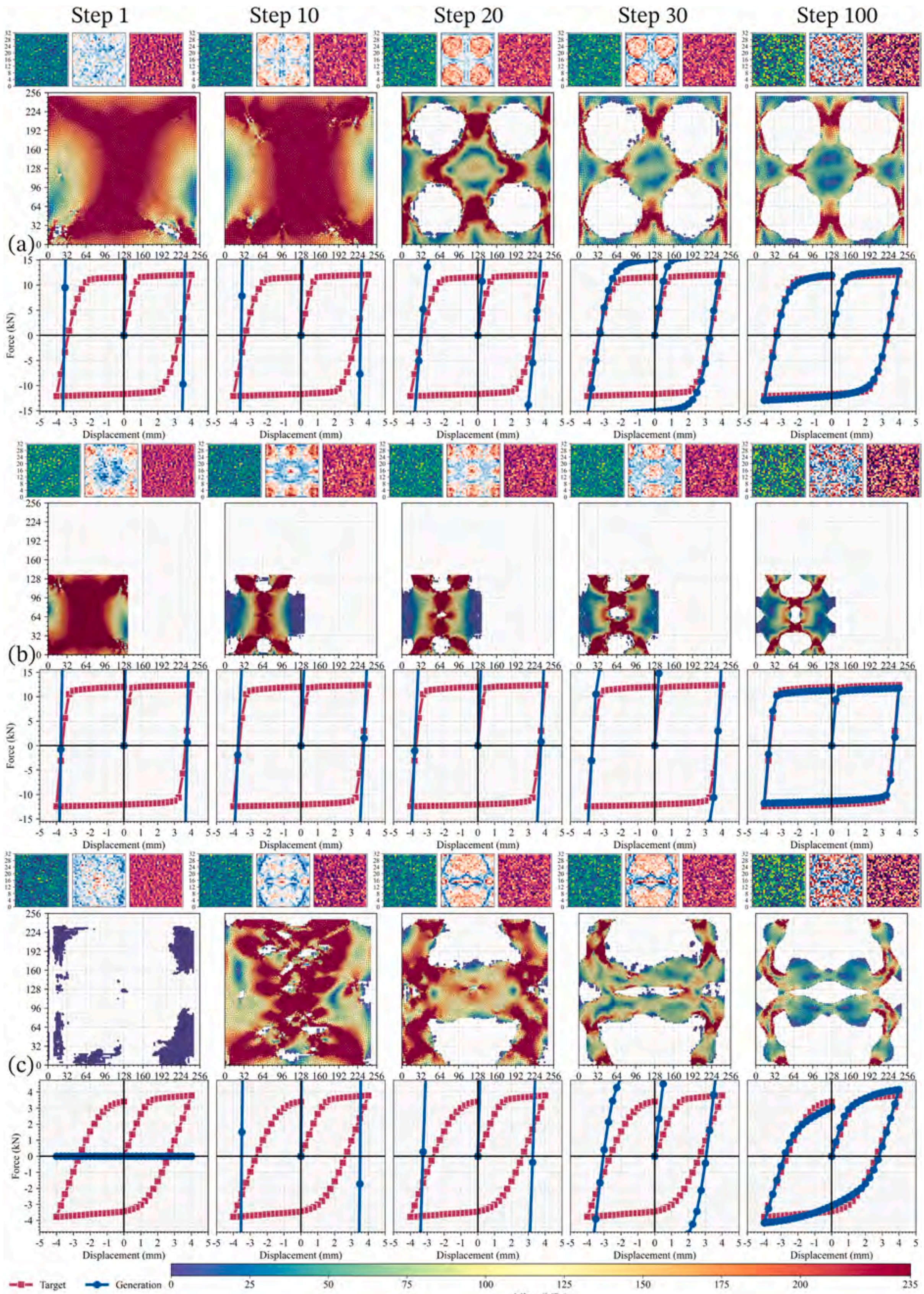

Fig. 13. The generation process of multiscale damping microstructures.

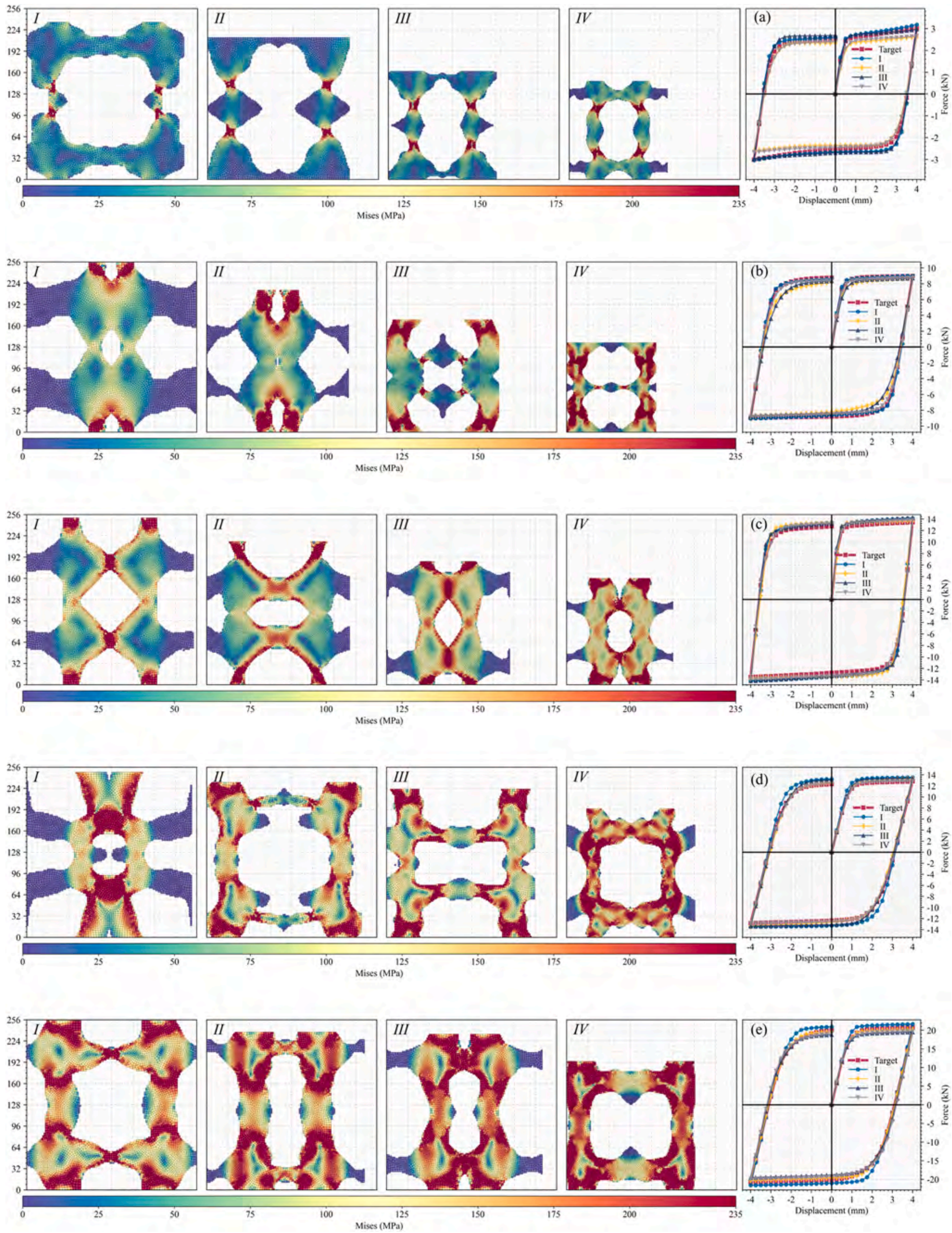

Fig. 14. The generation results of multiscale damping microstructures.

Table 4
Error statistics of mechanical performance for generated microstructures.

Figure		RMSE (kN)	MPAE (%)	$\text{Error}_{\text{yield force}}$ (%)	$\text{Error}_{\text{max force}}$ (%)	$\text{Error}_{\text{stiffness}}$ (%)	$\text{Error}_{\text{energy}}$ (%)
a	I	0.16	10.88	4.49	5.51	7.89	5.73
	II	0.25	13.25	9.52	12.87	24.48	9.50
	III	0.17	8.89	2.90	1.77	3.10	5.81
	IV	0.20	10.96	7.68	11.65	21.13	7.48
b	I	0.24	4.49	1.28	0.41	11.05	0.66
	II	0.68	8.44	6.03	2.92	1.39	7.68
	III	0.72	9.68	5.39	2.92	11.30	7.79
	IV	0.28	4.44	2.12	2.04	0.82	2.86
c	I	0.70	6.93	5.62	5.22	1.42	4.93
	II	0.74	7.79	4.03	1.92	10.17	3.41
	III	0.67	6.34	5.73	5.80	13.04	5.29
	IV	0.51	5.07	1.92	3.16	7.04	3.59
d	I	0.93	10.59	6.57	5.63	10.59	9.22
	II	0.45	6.69	4.86	4.10	2.40	0.48
	III	0.57	6.27	4.21	4.56	0.76	3.90
	IV	0.16	1.69	0.23	0.70	1.82	0.25
e	I	1.14	6.37	6.70	4.39	4.81	6.45
	II	0.40	2.50	1.51	2.74	1.53	1.16
	III	1.06	8.72	5.22	5.97	2.15	3.29
	IV	1.04	10.54	4.19	4.64	7.92	0.77

a significant difference in scale. In contrast, the structural scale in Fig. 13(a) is closer to that in Fig. 13(c), yet there is a substantial disparity in mechanical performance. Fig. 13 illustrates the generation process of damping microstructures, where the results generated at each timestep are decoded to obtain the microstructures. Subsequently, a finite element model is established for analysis to determine the mechanical properties at each timestep. During the first 30 timesteps, the generated results exhibit significant variability; however, after 30 timesteps, the results tend to converge. From the perspective of latent variables, the latent variable z_2 gradually reveals characteristics that are similar to those of the microstructures during the initial 30 timesteps. In contrast, at timestep 100, the visual representation of z_2 diverges considerably from the microstructure, yet the microstructure decoded at timestep 100 is evidently richer in detail. Even among microstructures with similar forms, substantial differences in mechanical performance remain.

5.4. The generation results of MultiDampGen

Five hysteresis curves were further selected as the generation conditions for MultiDampGen, and four physical scales were randomly chosen under each hysteresis curve, resulting in a total of 20 microstructures. As illustrated in Fig. 14, the red lines represent the mechanical performance demands for the generated microstructures, while the other lines correspond to the results of the hysteresis performance obtained from FEA for the generated microstructures.

In Fig. 14(a), the specified hysteresis performance demands indicate a maximum load capacity of 3.02 kN and a stiffness of 6.24 kN/mm, which represents relatively low values. However, the corresponding scales, designated as s of 1.84, 1.66, 1.27, and 1.15, imply that setting such a low load capacity while employing larger scales is relatively unreasonable. For instance, at s is 1.84, the corresponding dimension is 235.52 mm, and under shear deformation, the stress in this microstructure is primarily concentrated in the central region, with a significant portion of the area exhibiting low stress values. Other microstructures demonstrate similar characteristics, indicating that MultiDampGen is capable of generating microstructures that meet the specified conditions even under unreasonable settings. In Fig. 14(e), the structural load capacity demands are relatively high, leading to significant areas of the generated microstructure entering the plastic deformation regime under shear loading. This computational evidence confirms MultiDampGen's adaptable performance across varying load-bearing demands and energy-dissipating requirements. The structural generation capability can be constrained by mechanical performance criteria, allowing for the resultant microstructures to have their stress distribution patterns freely adjusted according to scale demands.

The error statistics for the generated microstructures are presented in Table 4, which summarizes the RMSE and Mean Absolute Percentage Error (MAPE) between the generated structures and the target demands, as well as the errors associated with yield force, maximum force, stiffness, and energy dissipation derived from the corresponding hysteresis curves. Overall, the errors for the generated microstructures are relatively low; however, a slightly larger stiffness error is observed under conditions of lower load capacity. This phenomenon can be attributed to the rather extreme conditions set and insufficient training samples. Crucially, excluding exceptional cases with irrational mechanical demands as exemplified in Table 4(a), the framework achieves remarkable precision in stiffness control and energy dissipation regulation, thereby validating the reliability of the proposed MultiDampGen framework across embedding physical significance in the generative domain.

For the microstructures generated in accordance with the hysteresis performance demands illustrated in Fig. 14, each microstructure is compared with the dataset containing 50,000 pre-existing microstructures. The scale ratio and SSIM are employed to assess the similarity between the generated microstructures and those in the dataset, as shown in Fig. 15. Specifically, a scale ratio of 1 indicates that the scales are identical, while an SSIM value of 1 signifies that the structural similarity is completely consistent. The experimental results demonstrate that all generated microstructures exhibit relatively low SSIM when compared to the structures in the dataset, with median values approximately around 0.2 and maximum values remaining below 0.7. Furthermore, when the scale ratio is equal to 1, the SSIM values decrease further, as shown in Fig. 15. This statistical evidence confirms that MultiDampGen successfully captured the intrinsic data distribution of damping microstructures rather than performing trivial selection from pre-existing mechanically compliant samples in the training set, thereby validating the efficacy of the framework in achieving genuine generative design capability as proposed in this study. Additionally, failure mode analysis and cyclic hysteresis degradation patterns of representative generated microstructures were experimentally validated in our related work, showing close agreement with FEA predictions (https://github.com/AshenOneme/DiT-Based-Microstructures-Design) and confirming their structural feasibility.

6. Conclusion and future work

This study presents a MultiDampGen framework for the efficient and high-quality generation of damping microstructures, which integrates three components: TopoFormer, RSV, and LDPM. Given specified

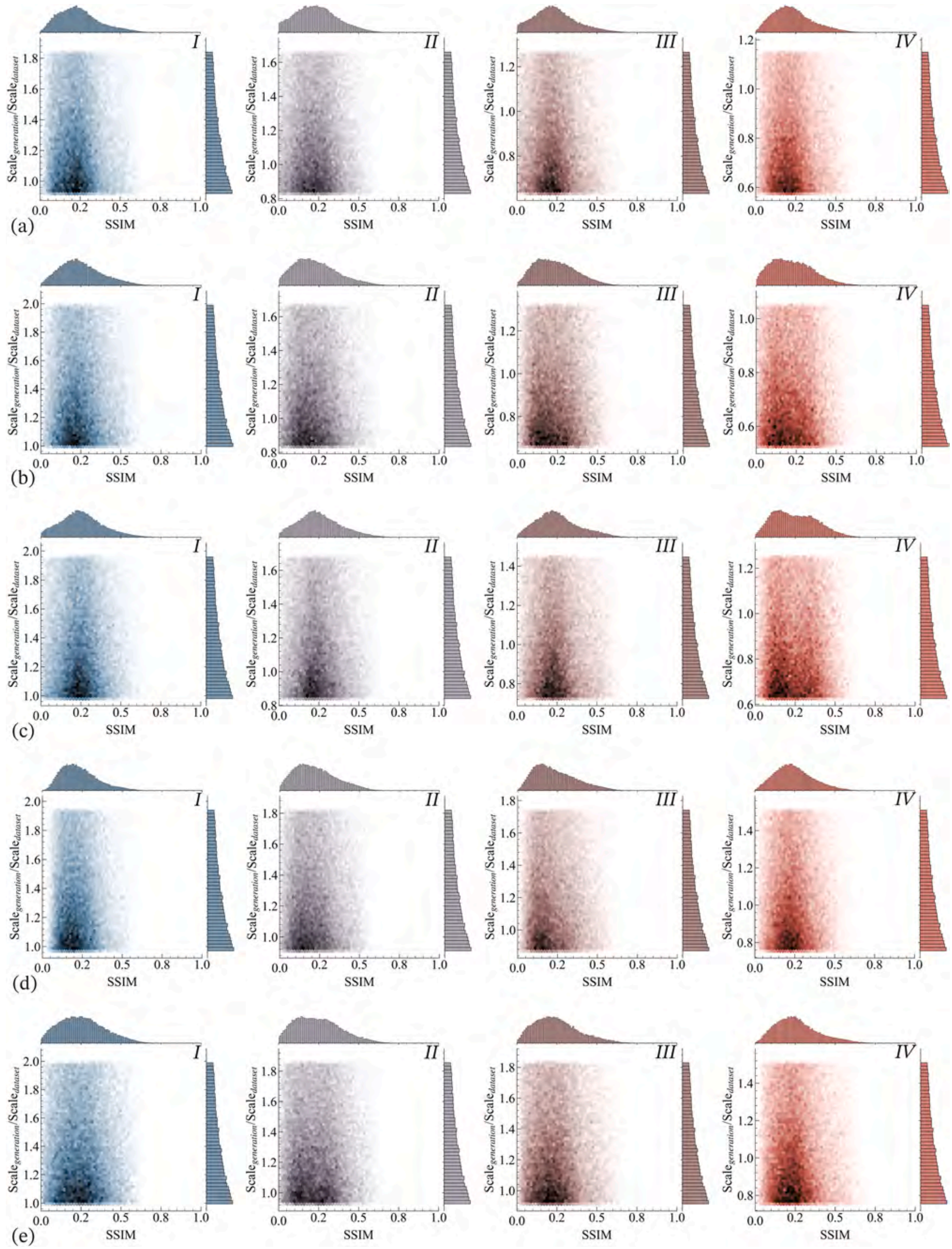

Fig. 15. Comparison of SSIM and scale.

mechanical performance demands and scales, this framework is capable of producing multiscale microstructures that meet hysteresis conditions. The main conclusions of this research are as follows:

1. TopoFormer operates as an attention-enhanced VAE architecture that achieves a quadratic reduction in generative complexity through latent feature compression of microstructures. Quantitative evaluations across 5000 microstructure samples demonstrate exceptional encoding fidelity, with SSIM reaching 0.998 between original and reconstructed microstructures.
2. The RSV functions as a discriminative evaluator in the generative pipeline, demonstrating remarkable hysteresis performance prediction accuracy while maintaining a small parameter count. Operating exclusively on latent variables and dimensional constraints as inputs, RSV achieves an equivalent prediction fidelity to FEA without requiring full-field microstructure reconstruction.
3. The MultiDampGen framework is capable of generating multiscale microstructures that satisfy hysteresis performance demands with a high degree of accuracy. The mechanical properties of the generated microstructures exhibit deviations from the target values that are consistently below 10 %. Notably, even under relatively unreasonable conditions, such as large scales with low load-bearing capacity and energy dissipation, the MultiDampGen framework is still able to produce satisfactory damping microstructures.

In the long term, the proposed MultiDampGen framework has the potential to shift the design paradigm of seismic energy dissipation systems from empirical, low-cycle reciprocating test-based approaches to intelligent, physics-integrated generative design processes. By enabling rapid exploration of vast and complex design spaces while ensuring compliance with mechanical performance requirements, this approach can significantly shorten development cycles and reduce both computational and material costs in retrofit and new construction projects. The openly released, physics-annotated dataset can serve as a standardized benchmark for research on shear-dominated damping microstructures, fostering reproducibility, transparency, and cross-institutional collaboration. Furthermore, the modular and multiscale architecture of MultiDampGen can be adapted to other engineering domains such as impact protection, vibration isolation, and advanced metamaterials, thereby broadening its relevance to various industrial domains. From a societal perspective, the widespread adoption of such AI-driven, physics-informed design tools could contribute to the development of safer, more resilient, and cost-effective urban infrastructure in seismic and other dynamic loading environments.

Although the proposed MultiDampGen framework performs well on high-performance computing platforms, its practical deployment in scenarios such as on-site structural monitoring systems or edge devices requires careful consideration of resource constraints. In many real-world applications, limitations in computational power, memory capacity, and energy supply may affect real-time generation and evaluation. While the latent diffusion approach significantly reduces generative complexity compared with pixel-space diffusion, further optimization is still necessary for low-resource environments. Possible strategies include model compression, knowledge distillation, mixed-precision inference, and the design of lightweight variants of TopoFormer and RSV. These directions will be investigated in future studies to enable efficient deployment and expand the practical applicability of the framework. However, some limitations remain in the current research. Due to the large size of the dataset, the finite element models were constructed using a voxel-based method that omits feature edges. This approach greatly improves the success rate of mesh generation but may reduce computational accuracy. Although the MultiDampGen framework can achieve precise designs for individual microstructures, the overall mechanical performance after periodic assembly, while influenced by the properties of individual units, still lacks a well-defined design model. Developing more robust multiscale design methodologies is therefore an important direction for future work.

CRediT authorship contribution statement

Weizhi Xu: Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Tianyang Zhang:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation. **Dongsheng Du:** Validation, Supervision, Funding acquisition, Conceptualization. **Shuguang Wang:** Visualization, Supervision, Project administration.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled "MultiDampGen: A Self-constraint Latent Diffusion Framework for Multiscale Energy-dissipating Microstructure Generation".

Acknowledgments

The authors are grateful for financial support from The National Natural Science Foundation of China (52208173) and The Natural Science Foundation of the Jiangsu Higher Education Institutions of China (22KJA560003).

Data availability

The relevant datasets and code are available at https://github.com/AshenOneme/MultiDampGen.

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